

Ex No: 2d

IMPLEMENTATION OF FORWARD CHAINING

Aim:

To implement forward Chaining.

Procedure:

1. Define the facts in the knowledge base.
2. Define the rules with conclusions and required conditions.
3. Initialize the inferred facts with the given facts.
4. Check each rule and infer a conclusion when all its conditions are satisfied.
5. Repeat the process until no new facts can be inferred.
6. Check whether the required goal is present in the inferred facts.
7. Display whether the goal can be achieved.

Program:

Facts: The knowledge base containing information about facts

```
facts = {  
    'a': True,  
    'b': True,  
    'c': False  
}
```

Rules: List of rules where each rule is a tuple (conclusion, conditions)

```
rules = [  
    ('d', ['a', 'b']), # d can be concluded if a and b are true  
    ('e', ['b', 'c']), # e can be concluded if b and c are true  
    ('f', ['d', 'e']) # f can be concluded if d and e are true  
]
```

Forward Chaining Function

```
def forward_chaining(facts, rules, goal):
```

```

inferred = set(facts.keys()) # Initialize inferred facts with the given facts
new_inferred = set(facts.keys()) # Start with the facts
while new_inferred:
    current_inferred = set() # Set to hold newly inferred facts
    for rule in rules:
        head, body = rule
        if head not in inferred and all(fact in inferred for fact in body):
            current_inferred.add(head)
            if current_inferred:
                inferred.update(current_inferred)
            new_inferred = current_inferred # Set new inferred facts for the next iteration
        else:
            new_inferred = set() # No new facts can be inferred

    return goal in inferred # Check if the goal is in the inferred facts

# Example Usage
goal = 'f'

if forward_chaining(facts, rules, goal):
    print(f"The goal '{goal}' can be achieved.")
else:
    print(f"The goal '{goal}' cannot be achieved.")

```

Output:

The goal 'f' can be achieved.

The goal 'e' cannot be achieved.

The goal 'd' can be achieved.

... The goal 'f' can be achieved.
The goal 'e' cannot be achieved.
The goal 'd' can be achieved.